

Technical Summary:

ConU: Conformal Uncertainty in Large Language Models with Correctness Coverage Guarantees

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1 Research Problem and Motivation

Large language models (LLMs) in open-ended natural language generation (NLG) can hallucinate facts while producing confident but incorrect responses, which undermines reliability in high-stakes usage. Uncertainty quantification (UQ) is required to support risk-aware decision-making and hallucination detection, but modern LLM access is often restricted to API-based black-box interaction that exposes only textual inputs and outputs. Existing UQ approaches frequently rely on white-box signals such as token likelihoods, embeddings, internal activations, or fine-tuning, which are unavailable or expensive under black-box constraints.

Conformal prediction (CP) can convert a heuristic uncertainty notion into statistically valid prediction sets with a user-specified error rate α , but prior CP adaptations for NLG have been constrained by multiple-choice reformulations, reliance on likelihood access, or inconsistent guarantees across settings. The core gap is a practical, black-box method that (i) produces calibrated prediction sets in the unbounded answer space of open-ended NLG, (ii) links the CP nonconformity score to correctness-aligned uncertainty conditions, and (iii) attains rigorous correctness coverage guarantees while maintaining small set sizes.

2 Related Work

2.1 Uncertainty Quantification in LLMs

White-box UQ methods utilize token-likelihoods or internal representations, and can suffer from calibration issues and substantial computational requirements. Black-box UQ for LLMs has emphasized sampling-based strategies, leveraging semantic similarity and dispersion among multiple generations to estimate uncertainty without logit access.

2.2 Conformal Prediction for LLMs

CP provides distribution-free coverage guarantees under exchangeability using a calibration set and a non-conformity score (NS). For LLMs, prior approaches have applied CP to multiple-choice question answering by converting NLG to fixed-option selection, which limits applicability to real open-ended outputs. Conformal language modeling calibrates a stopping rule using likelihoods to sample from the infinite output space, which is not practical for API-only models. LofreeCP addresses logit-free CP using diverse uncertainty signals but is reported to be sensitive to dataset and model settings and does not consistently guarantee correctness coverage.

2.3 Positioning of ConU

The study focuses on black-box LLMs in open-ended NLG and constructs a CP pipeline in which the NS is strictly aligned with the uncertainty condition of correct answers, enabling a robust conformal uncertainty criterion that calibrates prediction sets with correctness coverage guarantees across different models and datasets.

3 Dataset Construction

The empirical evaluation uses four free-form NLG datasets spanning general and medical domains. Dataset construction here refers to dataset selection, splits used for calibration and testing, and prompt formation for generation.

Table 1: Datasets, sources, and evaluation usage.

Dataset	Source and characteristics	Scale	Splits and usage in the study
CoQA	Conversational question answering with contextual information.	More than 127k question-answer pairs.	For UQ evaluation, 3,000 samples are randomly selected. For correctness coverage verification, the development set of 7,983 questions is used.
TriviaQA	Reading comprehension question answering.	Over 650k question-answer pairs.	For UQ evaluation, 3,000 samples are randomly selected. The same 3,000 samples are used for correctness coverage verification.
MedQA	Medical question answering collected from professional medical board exams. Described as MCQA dataset, evaluated here under free-form generation with semantic correctness.	Not specified in the paper.	Full validation set is used for correctness coverage verification. For UQ evaluation, 3,000 samples are randomly selected.
MedMCQA	Medical entrance exam questions, described as large-scale MCQA dataset, evaluated here under free-form generation with semantic correctness.	Not specified in the paper.	Full validation set is used for correctness coverage verification. For UQ evaluation, 3,000 samples are randomly selected.

Prompting and preprocessing. For CoQA, the prompt concatenates contextual information with the question. For TriviaQA and MedMCQA, a fixed few-shot template uses 5 question-answer pairs and is combined with the current question. For MedQA, a few-shot template uses 3 question-answer pairs. Additional preprocessing details are not specified in the paper.

4 Query Protocol and Task Definitions

4.1 Input, output, and generation protocol

Each sample is a prompt (question) x_i with a reference answer y_i^* used only for evaluation and calibration. The protocol generates (i) a most likely generation $\hat{y}_i$ for correctness evaluation, and (ii) a sampled candidate set $\{\hat{y}_m^{(i)}\}_{m=1}^M$ for black-box UQ and prediction-set construction. Candidate generations are obtained by multinomial sampling, while the most likely generation is obtained by beam search.

4.2 Task definitions and success standards

Task A: Black-box uncertainty quantification. UQ is evaluated as failure prediction, namely whether an uncertainty score separates correct versus incorrect generations. The success criterion is higher AUROC, where higher AUROC indicates stronger separability.

Task B: Conformal correctness coverage in open-ended NLG. For a user-specified error rate α , the goal is a calibrated prediction set $P(x_{\text{test}})$ that contains at least one correct answer with probability at least $1 - \alpha$, while keeping the average set size small for efficiency.

4.3 Correctness and semantic equivalence

Correctness and equivalence are operationalized by sentence semantic similarity computed using a cross-encoder with DistillRoBERTa backbone. A strict similarity threshold of 0.7 is used to determine correctness and semantic equivalence between a generation and the reference answer.

4.4 Assumptions for coverage guarantees

Two assumptions are required: (1) at least one candidate generation in $\{\hat{y}_m^{(\text{test})}\}_{m=1}^M$ satisfies the correctness criterion, and (2) calibration and test samples are exchangeable, which is the standard requirement for CP validity.

4.5 Conformal quantile and prediction set construction

The conformal threshold $\hat{q}$ is computed as the $(1 - \alpha)$ quantile of calibration nonconformity scores, and the prediction set includes candidate generations whose nonconformity score is below $\hat{q}$.

$$\hat{q} = \inf \left\{ q : \frac{|\{i : r_i \leq q\}|}{N} \geq \frac{\lceil (N+1)(1-\alpha) \rceil}{N} \right\} = r_{\lceil (N+1)(1-\alpha) \rceil}. \quad (4)$$

Purpose. Calibrates an uncertainty threshold from a fresh calibration set to control the correctness coverage error rate at most α .

Symbols. N is the number of calibration samples; r_i is the nonconformity score for calibration sample i ; $\alpha \in (0, 1)$ is the user-specified upper bound on error rate; $\lceil \cdot \rceil$ is the ceiling operator; $\hat{q}$ is the conformal uncertainty criterion (threshold).

Intuition. Correct answers in calibration define a distribution of correctness-aligned uncertainty. The threshold is set so that only an α fraction of calibration scores can exceed it, up to finite-sample correction.

Usage. After computing $\{r_i\}_{i=1}^N$, sorting them, and selecting the indexed quantile, $\hat{q}$ is used to filter candidate generations for each test prompt.

$$P(x_{\text{test}}) = \left\{ \hat{y}_j^{(\text{test})} : r\left(x_{\text{test}}, \hat{y}_j^{(\text{test})}\right) \leq \hat{q} \right\}. \quad (5)$$

Purpose. Constructs a calibrated prediction set for a test prompt by selecting candidate generations that satisfy the calibrated uncertainty condition.

Symbols. x_{test} is a test prompt; $\hat{y}_j^{(\text{test})}$ is a sampled candidate generation; $r(x, \hat{y})$ is the test-time nonconformity score; $\hat{q}$ is the conformal threshold; $P(x_{\text{test}})$ is the prediction set.

Intuition. If correctness-aligned generations tend to have low nonconformity, filtering by $\hat{q}$ retains a small subset enriched for correct answers.

Usage. Candidate generations are sampled for each test prompt, scored with $r(\cdot, \cdot)$, and filtered using $\hat{q}$ to obtain $P(x_{\text{test}})$.

$$\mathbb{P}(y_{\text{test}}^* \in P(x_{\text{test}})) \geq 1 - \alpha. \quad (6)$$

Purpose. States the formal correctness coverage guarantee of the calibrated prediction sets under exchangeability and the stated candidate-set assumption.

Symbols. y_{test}^* is the reference correct answer for the test prompt; $P(x_{\text{test}})$ is the prediction set; α is the user-specified error rate.

Intuition. Under exchangeability, the rank of the test nonconformity score among calibration scores is uniformly distributed, yielding a distribution-free coverage lower bound.

Usage. This guarantee justifies using $P(x_{\text{test}})$ as a correctness-preserving candidate pool for downstream selection or abstention.

5 Modeling Approach

5.1 Overview

The approach addresses two issues: black-box UQ from text-only access and rigorous correctness coverage in open-ended NLG. The pipeline is: (i) sample multiple candidate generations for each prompt, (ii) perform semantic clustering to identify distinct meanings, (iii) compute a generation-level uncertainty score based on self-consistency and cross-cluster semantic consistency, (iv) define CP nonconformity scores on calibration data by aligning with correctness, (v) compute a conformal threshold and construct prediction sets for test prompts, and (vi) optionally perform selective prediction by choosing the lowest-uncertainty element in each prediction set.

5.2 Sampling and semantic clustering

For each prompt, M candidate generations are sampled from the model output space. Semantic clustering groups candidates into K distinct semantic clusters, where cluster k contains V_k generations and $\sum_{k=1}^K V_k = M$. The frequency of a semantic cluster is used as a proxy for reliability following self-consistency theory.

5.3 Generation-level uncertainty score

$$U(\hat{y}_m^{(i)}) = 1 - \lambda \cdot F(\hat{y}_m^{(i)}) - (1 - \lambda) \cdot \frac{1}{K} \sum_{k=1}^K S(\hat{y}_m^{(i)}, \hat{y}_k^{(i)}) F(\hat{y}_k^{(i)}). \quad (1)$$

Purpose. Defines a sampling-based black-box uncertainty score for each candidate generation, combining semantic self-consistency and cross-semantic agreement.

Symbols. $\hat{y}_m^{(i)}$ is the m -th sampled generation for prompt x_i ; K is the number of distinct semantic clusters; $\hat{y}_k^{(i)}$ denotes any representative generation in cluster k ; $F(\hat{y}) \in [0, 1]$ is the frequency of the semantic cluster conveyed by $\hat{y}$, defined as V_k/M for cluster size V_k ; $S(\cdot, \cdot)$ is a semantic similarity score computed by a

cross-encoder model; $\lambda \in [0, 1]$ is a coefficient controlling the trade-off between frequency and cross-cluster consistency.

Intuition. A generation is reliable when its meaning appears frequently among samples and when it is semantically consistent with other frequent meanings. The score is an uncertainty measure, so higher frequency and stronger agreement reduce uncertainty.

Usage. After sampling and clustering, $F(\cdot)$ and $S(\cdot, \cdot)$ are computed among candidates, and $U(\hat{y}_m^{(i)})$ is used as the base uncertainty signal for both UQ evaluation and CP calibration.

5.4 Query-level uncertainty for the sampling process

$$U\left(\{\hat{y}_m^{(i)}\}_{m=1}^M \mid x_i\right) = 1 - \lambda \cdot F(\hat{y}_i^{\text{mst}}) - (1 - \lambda) \cdot \frac{1}{K} \sum_{k=1}^K S(\hat{y}_i^{\text{mst}}, \hat{y}_k^{(i)}) F(\hat{y}_k^{(i)}). \quad (2)$$

Purpose. Quantifies the overall uncertainty of the query-response process by using the most frequent semantic as a reference object.

Symbols. $\hat{y}_i^{\text{mst}}$ is any generation in the largest semantic cluster among $\{\hat{y}_m^{(i)}\}_{m=1}^M$, interpreted as the model’s most confident semantic; other symbols follow Eq. (1).

Intuition. The most frequent semantic represents the dominant belief under sampling. Uncertainty increases when this dominant semantic is rare or deviates from other frequent semantics.

Usage. Used to summarize uncertainty at the prompt level and to support the interpretation of black-box UQ behavior.

5.5 Correctness-aligned nonconformity score for calibration

$$r_i = r(x_i, y_i^*) = U\left(\arg \max_{\hat{y}_j^{(i)}} S(\hat{y}_j^{(i)}, y_i^*) E(\hat{y}_j^{(i)}, y_i^*)\right). \quad (3)$$

Purpose. Defines the CP nonconformity score on calibration data by selecting a correctness-equivalent generation and using its uncertainty as the calibration signal.

Symbols. r_i is the nonconformity score for calibration sample i ; y_i^* is the reference answer; $E(\cdot, \cdot) \in \{0, 1\}$ is an indicator of semantic equivalence, where $E = 1$ indicates equivalence under the correctness criterion; $S(\cdot, \cdot)$ is the semantic similarity score; $U(\cdot)$ is the uncertainty score in Eq. (1).

Intuition. Calibration must reflect uncertainty conditions of correct answers. Selecting the most similar generation among those deemed semantically equivalent to the reference tightly couples r_i with correctness.

Usage. For each calibration prompt, candidates are sampled, filtered by equivalence to the reference using $E(\cdot, \cdot)$, the most similar one is selected, and its uncertainty becomes r_i , which is then used to compute $\hat{q}$ in Eq. (4).

5.6 Models and components used

Generator LLMs. Seven instruction-tuned LLMs are evaluated: GPT-3.5-turbo, LLaMA-2-7B-Chat, Mistral-7B-Instruct-v0.3, Llama-3-8B-Instruct, Vicuna-13B-v1.5, LLaMA-2-13B-Chat, and LLaMA-3-70B-Instruct. Open-source models use default generation configurations and checkpoints from HuggingFace.

Semantic similarity model. A cross-encoder with DistillRoBERTa backbone computes semantic similarity for clustering, correctness evaluation, and equivalence testing.

Table 2: Training and evaluation setup. Items not stated are marked as Not specified in the paper.

Category	Details
Task type	Open-ended NLG (free-form QA style) with black-box UQ and conformal prediction sets.
Most likely generation	Beam search for correctness evaluation.
Candidate generations	Multinomial sampling for UQ and CP prediction sets.
Candidate set size M	5 generations per prompt for UQ evaluation; 10 generations per prompt for correctness coverage verification.
Max generation length	128 tokens.
Temperature	1.0.
Trade-off coefficient λ	0.5.
Calibration:test split	1:10 by default.
Correctness threshold	Sentence semantic similarity threshold of 0.7.
Optimizer, learning rate, batch size, epochs	Not specified in the paper.
Fine-tuning or additional training	None required for the proposed method; no task-specific fine-tuning is used for selective prediction.
Hardware, framework versions, cost	Not specified in the paper.

6 Empirical Results

6.1 Baselines and evaluation metrics

Eight baselines are compared: four white-box methods, Predictive Entropy (PE), Length-normalized Predictive Entropy (LNPE), Semantic Entropy (SE), and Shift Attention to Relevance (SAR); and four black-box methods, Lexical Similarity (LS), Number of Semantic Sets (NumSet), Eccentricity (Ecc), and Degree Matrix (Deg). UQ performance is measured by AUROC under failure prediction. Conformal performance is evaluated by correctness coverage rate under various α and by average prediction set size.

6.2 Uncertainty quantification performance (AUROC)

Table 1 reports AUROC on five instruction-tuned LLMs across four datasets. ConU yields the highest average AUROC on TriviaQA (0.8224), improving over the strongest baseline Deg (0.8167) and SAR (0.7883). On MedQA, ConU achieves 0.6524 average AUROC, exceeding Deg (0.6443) and SAR (0.6214). On MedMCQA, ConU achieves 0.6741 average AUROC, exceeding Deg (0.6511) and NumSet (0.6503). On CoQA, ConU average AUROC is 0.7530, slightly lower than Deg (0.7552), indicating that ConU is not the top method on every dataset-model combination. The reported gains in medical datasets include up to 2.4% AUROC improvement on MedMCQA and 1.29% AUROC improvement on MedQA, highlighting measurable but bounded margins.

6.3 Correctness coverage guarantees

With $\alpha = 0.1$, Table 2 shows correctness coverage rates at or above 90% across seven LLMs and four datasets, satisfying the target lower bound implied by Eq. (6). Examples include 100.00% on MedQA for LLaMA-2-7B-Chat, 94.27% on TriviaQA for LLaMA-3-8B-Instruct, and coverage near 90% for multiple model-dataset pairs such as 90.36% on MedMCQA for GPT-3.5-turbo and 90.39% on MedMCQA for Mistral-7B-Instruct-v0.3.

6.4 Prediction efficiency (average set size)

Table 3 reports small average prediction set sizes under $\alpha = 0.1$, supporting practical usability. For LLaMA-3-70B-Instruct, the average set size is 1.03 on TriviaQA and 1.60 on MedMCQA, indicating near-singleton sets. For smaller models, average set sizes are typically between roughly 2 and 4, with larger values in medical datasets for some models, such as 4.28 on MedQA for LLaMA-2-7B-Chat. These results demonstrate an explicit efficiency advantage: prediction sets remain small while maintaining coverage.

6.5 Selective prediction utility

Selective prediction selects the generation with the lowest uncertainty score within each calibrated prediction set and evaluates accuracy. Table 4 shows consistent accuracy gains. Examples include MedQA on LLaMA-3-8B-Instruct improving from 66.29% to 70.59%, TriviaQA on Mistral-7B-Instruct-v0.3 improving from 79.04% to 81.45%, and CoQA on LLaMA-2-7B-Chat improving from 73.00% to 75.53%. Improvements are smaller in some cases, such as TriviaQA on Vicuna-13B-v1.5 improving from 78.40% to 78.80%, indicating dependence on the quality of the uncertainty score and the presence of empty prediction sets.

6.6 Ablation studies and robustness

Number of sampled generations. Increasing the number of sampled generations improves UQ performance. With only 4 generations, the method achieves AUROC 0.8082 on TriviaQA with LLaMA-3-8B-Instruct, demonstrating generation efficiency.

Calibration size ratio. Varying the calibration:test ratio preserves a strict lower bound on coverage when prediction sets are constructed with the conformal uncertainty criterion, indicating robustness to calibration size changes.

Stricter error rates. Coverage robustness is further supported at $\alpha = 0.05$ and $\alpha = 0.01$. For $\alpha = 0.05$, coverage rates in Table 5 are approximately 95% or higher across models and datasets, including 100.00% on MedQA for LLaMA-2-7B-Chat. For $\alpha = 0.01$, Table 6 reports coverage near 99% to 100% across settings, such as 99.93% on TriviaQA and 100.00% on MedQA for LLaMA-2-7B-Chat.

7 Summary

A black-box, sampling-based uncertainty measure (ConU) is developed for open-ended NLG by combining (i) semantic frequency as a self-consistency proxy and (ii) cross-semantic weighted consistency computed via cross-encoder similarity. A conformal uncertainty criterion is constructed by defining calibration non-conformity scores strictly from correctness-aligned uncertainty conditions, enabling calibrated prediction sets with rigorous correctness coverage guarantees under exchangeability and a candidate-set correctness assumption.

Empirically, ConU generally improves UQ performance measured by AUROC across diverse LLMs and datasets, with notable gains on TriviaQA and medical datasets, while showing a small disadvantage on CoQA average relative to Deg. The conformal pipeline achieves strict control of correctness coverage under multiple user-specified error rates, with small average prediction set sizes, and supports selective prediction that improves accuracy without task-specific fine-tuning or architectural modification.

Limitations and future directions (as stated)

Three limitations are identified: (1) a criterion to verify whether a correct answer has been sampled from the output space is required for real-world deployment, (2) findings are limited to four datasets and extension

to other NLG tasks such as document summarization is planned, and (3) the method targets exchangeable settings and an extension to non-exchangeability scenarios is proposed.